

โครงสร้างตาราง 4 ตาราง (Publication-ready)

Table 1 — Regional Mean Performance (แนวตั้ง, A4) ตารางหลักที่สำคัญที่สุด แสดง Raw CMIP6 (แดง) และ BC/QDM (น้ำเงิน) แยกกันชัดเจน ทุก N=1–6 พร้อม mean $\pm$ SD ครบ 4 metrics + Spread แถว N*=5 แรเงาเขียวอ่อน เส้นบนตาราง/ล่างตารางหนา 2 เส้น (journal standard)

Table 2 — Bootstrap CI for N* (แนวตั้ง, A4) แสดงความน่าเชื่อถือของ N* ด้วย 95% CI และ bootstrap statistics ทั้ง 4 metrics พร้อม Table 2b แสดง CI ของ performance ทุก N รองรับการอ้างอิง Efron & Tibshirani (1993)

Table 3 — Per-Station Saturation (แนวนอน, Landscape) ตารางใหญ่ 12 สถานี $\times$ 4 metrics แสดง N*, @N*, @N=6 ครบ เซลล์เหลือง = S8/S9 ที่ KGE อิมตัวเร็ว (N*=4 และ 3) ซึ่งเป็น finding ที่น่าสนใจสำหรับ discussion

Table 4 — Data Validation (แนวตั้ง, A4) สรุปความถูกต้องของข้อมูลและ % missing values ก่อน/หลัง QDM เหมาะสำหรับ Supplementary หรือ Appendix ใน Methods section

Table 1. Regional mean performance metrics of raw CMIP6 and bias-corrected (QDM) ensemble simulations as a function of ensemble size (N) for daily rainfall at 12 stations in Prachuap Khiri Khan Province, Thailand (1981–2014). Values represent the mean $\pm$ 1 SD across all C(6,N) sub-ensemble combinations. Saturation point N* is indicated by ★.

Dataset	N	C(6,N)	Performance metric (mean $\pm$ SD)				Spread (mm d ⁻¹)
			RMSE (mm d ⁻¹)	KGE	NSE	r	
Raw CMIP6	1	6	10.00 $\pm$ 0.65	−0.014 $\pm$ 0.019	−0.716 $\pm$ 0.253	0.094 $\pm$ 0.022	0.00
	2	15	9.05 $\pm$ 0.23	0.010 $\pm$ 0.012	−0.340 $\pm$ 0.083	0.120 $\pm$ 0.014	2.97
	3	20	8.70 $\pm$ 0.12	0.008 $\pm$ 0.009	−0.215 $\pm$ 0.043	0.137 $\pm$ 0.010	3.38
	4	15	8.51 $\pm$ 0.08	0.005 $\pm$ 0.007	−0.152 $\pm$ 0.025	0.149 $\pm$ 0.007	3.60
	5 ★	6	8.40 $\pm$ 0.05	0.002 $\pm$ 0.005	−0.115 $\pm$ 0.015	0.158 $\pm$ 0.005	3.75
	6	1	8.32 $\pm$ —	0.000 $\pm$ —	−0.090 $\pm$ —	0.165 $\pm$ —	3.86
BC (QDM)	1	6	10.14 $\pm$ 0.05	0.098 $\pm$ 0.008	−0.557 $\pm$ 0.016	0.113 $\pm$ 0.008	0.00
	2	15	9.01 $\pm$ 0.03	0.075 $\pm$ 0.006	−0.233 $\pm$ 0.010	0.145 $\pm$ 0.006	3.03
	3	20	8.60 $\pm$ 0.02	0.055 $\pm$ 0.005	−0.125 $\pm$ 0.007	0.164 $\pm$ 0.005	3.52
	4	15	8.39 $\pm$ 0.02	0.043 $\pm$ 0.004	−0.071 $\pm$ 0.005	0.178 $\pm$ 0.004	3.81
	5 ★	6	8.26 $\pm$ 0.01	0.035 $\pm$ 0.003	−0.039 $\pm$ 0.003	0.189 $\pm$ 0.003	4.02
	6	1	8.18 $\pm$ —	0.029 $\pm$ —	−0.017 $\pm$ —	0.197 $\pm$ —	4.18
N* (RMSE / NSE / KGE / r)			Raw: 5 / 5 / 6 / 6		BC: 5 / 5 / 6 / 6		

C(6,N) = number of exhaustive sub-ensemble combinations; SD = standard deviation across all combinations; Spread = inter-model spread (mm d⁻¹); ★ = saturation point N* (first N where marginal gain < 5% of total gain); — = single combination (SD undefined); RMSE = root-mean-square error; KGE = Kling–Gupta efficiency; NSE = Nash–Sutcliffe efficiency; r = Pearson correlation coefficient; QDM = quantile delta mapping.

Table 2. Bootstrap confidence intervals (95%) for the ensemble saturation point N^* and regional mean performance at N^* for bias-corrected (QDM) simulations. Bootstrap resampling was conducted with replacement ($n = 999$ iterations) following Efron and Tibshirani (1993).

Metric	N^* (point est.)	Bootstrap mean N^*	Bootstrap SD	95% CI	Perf. at N^*	Perf. at $N = 6$	Residual gain (%) ^a
RMSE (mm d ⁻¹)	5	5.02	0.13	[5, 5]	8.263	8.175	1.1
KGE	6	5.99	0.09	[6, 6]	0.029	0.029	0.0
NSE	5	5.00	0.03	[5, 5]	-0.039	-0.017	—
r	6	6.00	0.00	[6, 6]	0.197	0.197	0.0

^a Residual gain = additional percentage improvement achievable beyond N^* when adding remaining models up to $N = 6$; — = negative metric (NSE), percentage not applicable.

Table 2b. Bootstrap 95% confidence intervals for regional mean performance at each ensemble size N (BC/QDM).

N	RMSE (mm d ⁻¹)			KGE			NSE			r		
	Mean	CI lo	CI hi	Mean	CI lo	CI hi	Mean	CI lo	CI hi	Mean	CI lo	CI hi
1	10.136	10.095	10.168	0.098	0.095	0.102	-0.557	-0.568	-0.546	0.113	0.109	0.117
2	9.012	8.995	9.027	0.075	0.072	0.077	-0.233	-0.238	-0.228	0.145	0.142	0.148
3	8.604	8.594	8.614	0.055	0.053	0.057	-0.125	-0.128	-0.122	0.164	0.162	0.167
4	8.392	8.383	8.401	0.043	0.041	0.045	-0.071	-0.074	-0.068	0.178	0.176	0.180
5 ★	8.263	8.255	8.272	0.035	0.033	0.037	-0.039	-0.041	-0.036	0.189	0.186	0.191
6	8.175	8.175	8.175	0.029	0.029	0.029	-0.017	-0.017	-0.017	0.197	0.197	0.197

★ = saturation point N^* ; CI lo/hi = lower/upper bound of 95% bootstrap percentile confidence interval.

Table 3. Station-level saturation points (N^*) and performance at saturation for bias-corrected (QDM) ensemble simulations at 12 rainfall stations in Prachuap Khiri Khan Province (1981–2014). N^* is defined as the first ensemble size where marginal improvement falls below 5% of total gain. Bold values indicate $N^* \neq 6$ (early saturation).

Station	RMSE (mm d ⁻¹)			KGE			NSE			r		
	N^*	@ N^*	@ $N=6$	N^*	@ N^*	@ $N=6$	N^*	@ N^*	@ $N=6$	N^*	@ N^*	@ $N=6$
S1 (500001)	5	9.145	9.031	6	−0.033	−0.033	5	−0.117	−0.089	6	0.126	0.126
S2 (500002)	5	10.351	10.243	6	0.004	0.004	5	−0.066	−0.044	6	0.173	0.173
S3 (500003)	5	10.151	10.024	6	−0.047	−0.047	5	−0.122	−0.094	6	0.114	0.114
S4 (500004)	5	9.628	9.520	6	−0.047	−0.047	5	−0.112	−0.087	6	0.118	0.118
S5 (500005)	5	6.711	6.661	6	0.019	0.019	5	−0.011	0.004	6	0.212	0.212
S6 (500006)	5	6.995	6.948	6	−0.005	−0.005	5	−0.010	0.003	6	0.200	0.200
S7 (500007)	5	6.642	6.592	6	−0.002	−0.002	5	−0.014	0.001	6	0.200	0.200
S8 (500008)	5	3.706	3.663	4	0.350	0.352	5	0.178	0.197	6	0.470	0.470
S9 (500009)	5	3.883	3.839	3	0.315	0.317	5	0.149	0.168	6	0.442	0.442
S10 (500201)	5	11.375	11.244	6	−0.081	−0.081	5	−0.120	−0.095	6	0.092	0.092
S11 (500202)	5	10.527	10.409	6	−0.081	−0.081	5	−0.113	−0.089	6	0.096	0.096
S12 (500301)	5	10.040	9.926	6	−0.051	−0.051	5	−0.107	−0.082	6	0.119	0.119
Regional N^*	5 [95% CI: 5–5]			6 [95% CI: 6–6]			5 [95% CI: 5–5]			6 [95% CI: 6–6]		

@ N^* = metric value at saturation; @ $N=6$ = metric value at full ensemble; highlighted cells = early KGE saturation (S8: $N^*=4$, S9: $N^*=3$); green shading = $N^*=5$, dark green = $N^*=6$.

Table 4. Data validation summary for observed and CMIP6 model precipitation datasets used in ensemble saturation analysis, Prachuap Khiri Khan Province (1981–2014).

Check	Result	Status
Common temporal period	1981–2014	PASS
Common daily overlap	12,410 days	PASS
Date continuity gap	0.0%	PASS
Data errors / warnings	None	PASS

Table 4b. Missing value percentage by dataset before and after quantile delta mapping (QDM) bias correction.

Model	Raw CMIP6 (%)	BC/QDM (%)
Observed	0.00	—
ACCESS-ESM1-5	0.00	0.00
CESM2	0.00	0.00
CanESM5	0.00	0.00
EC-Earth3	0.22	0.00
MIROC6	0.00	0.00
MRI-ESM2-0	0.00	0.00

Highlighted row = EC-Earth3 had 0.22% missing values in raw data (27 days); missing values were infilled during QDM bias correction. All other datasets complete (0.00% missing). — = not applicable.